THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 9: **Light - Reflection and Refraction** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The refractive index of glass is 1.5 and of water is 1.33. Light travels fastest in 1

(a) glass **(b)** water **(c)** both equally **(d)** neither

2. The magnification of a lens is +3. The image is 1

(a) real and magnified **(b)** virtual and diminished

(c) virtual, erect and three times larger **(d)** real and inverted

3. The magnification for an object at 10 cm in front of a concave mirror of f = 20 cm is 1

(a) +2 **(b)** -2 **(c)** +0.5 **(d)** -0.5

4. A ray of light passes undeviated through a medium when it is incident 1

(a) along the surface **(b)** at 45 degrees

(c) along the normal **(d)** at 60 degrees

5. A concave lens has f = -30 cm. An object is at 30 cm. The image forms at 1

(a) 60 cm on the other side **(b)** 15 cm on the other side

(c) 30 cm on the same side **(d)** 15 cm on the same side

6. The image of an object in a mirror is of the same size and is erect. The mirror is 1

(a) plane **(b)** convex **(c)** concave **(d)** cylindrical

7. Which of these has the largest lateral displacement in a slab, for the same incidence angle? 1

(a) All the same **(b)** Slab of air **(c)** Thicker slab **(d)** Thinner slab

8. The power of a combination of lenses of +3 D and -1 D in contact is 1

(a) -2 D **(b)** +4 D **(c)** +2 D **(d)** +3 D

9. A mirror gives a virtual, erect and magnified image. The mirror is 1

(a) concave, with the object between P and F **(b)** plane with the object anywhere

(c) concave with the object at C **(d)** convex with the object at infinity

10. The optical density of a medium is higher if 1

(a) the speed of light in it is higher **(b)** it is lighter

(c) it is less dense physically **(d)** the speed of light in it is lower

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A ray enters glass of n = 1.5 at 30 degrees. Find sin r. 2

12. Find the focal length of a lens of power +4 D and tell whether it is converging or diverging. 2

13. A student wants a magnified virtual image of a small object using a concave mirror of f = 10 cm. State the range of object distances and justify. 2

14. Convert the power -2.5 D into focal length and name the lens. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A concave mirror produces three times magnified real image of an object at 10 cm. Find the image distance and focal length. 3

16. Two lenses of f = +25 cm and f = -40 cm are placed in contact. Find the net power. 3

17. An object is held at a distance of 25 cm from a concave mirror of focal length 20 cm. Find the image distance and nature. 3

OR

Light enters from air into glass (n = 1.5). If the angle of incidence is 60 degrees, find sin r and the speed of light in glass. 3

18. An object is placed 8 cm from a convex lens of f = 12 cm. Find the position, magnification and nature. 3

OR

The power of a lens is -4 D. Find the focal length and the image distance for an object 50 cm away. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student gets a sharp image of a candle on a screen kept 60 cm from a lens when the candle is 30 cm from the lens. Find the focal length, nature of the lens, magnification and the image size if the candle is 2 cm tall. 5

OR

A thin lens of power +5 D is combined with a lens of power -3 D. Find the equivalent power and focal length. An object is placed 100 cm from the combination. Find the image position. 5

20. Explain the refraction of light through a rectangular glass slab. A ray enters at 30 degrees and glass has n = $\sqrt{3}$. Find the angle of refraction and angle of emergence. 5

OR

Distinguish between a real image and a virtual image and state how each is formed by a concave mirror. Give the condition and an application for each. 5

END OF QUESTION PAPER — Total: 40 Marks

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